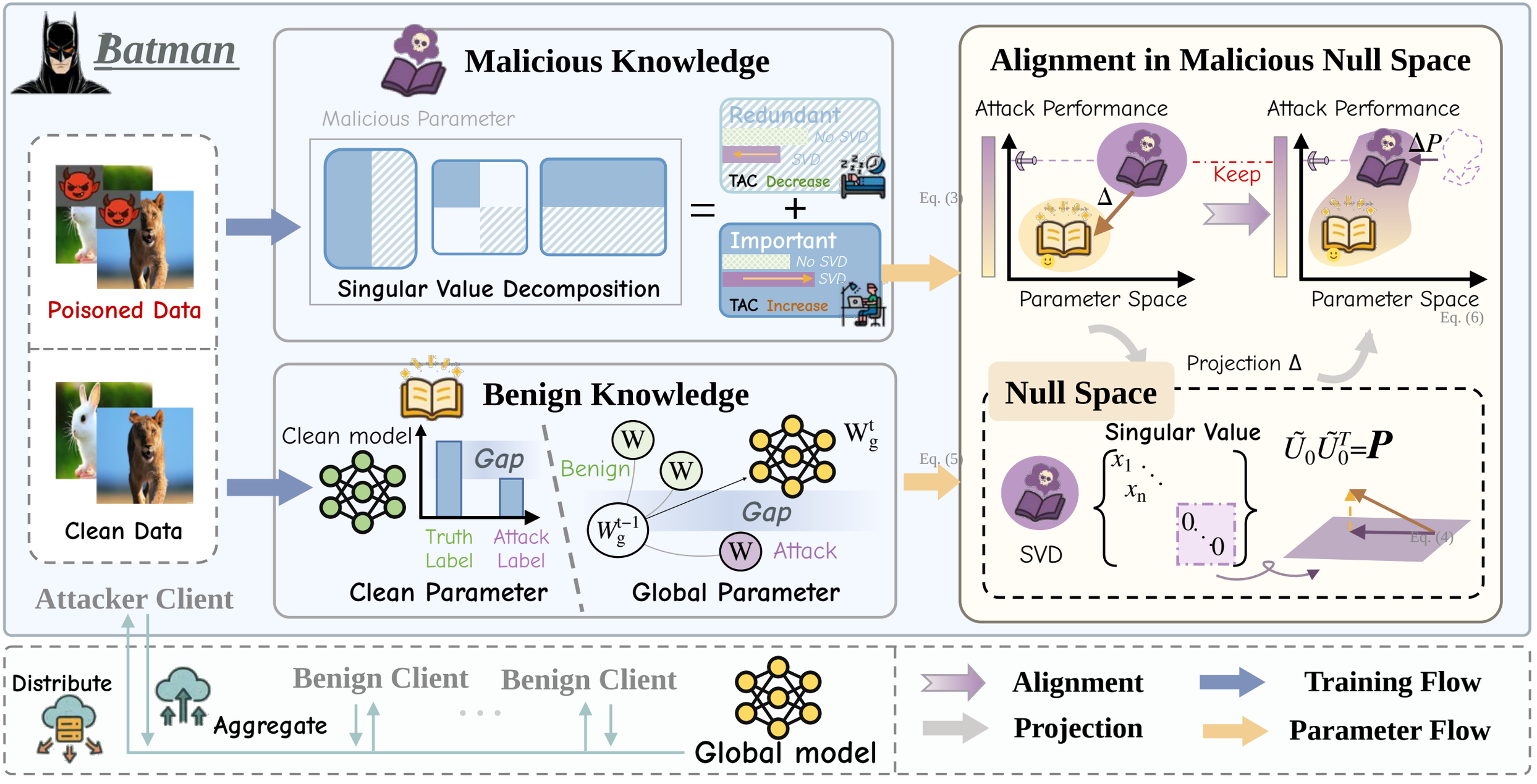

Batman
Malicious Knowledge
Malicious Parameter
Singular Value Decomposition
Redundant
No SVD
SVD
TAC Decrease
Important
No SVD
SVD
TAC Increase
Poisoned Data
Clean Data
Attacker Client
Benign Knowledge
Clean model
Gap
Truth Label
Attack Label
Clean Parameter
Benign
Gap
Attack
Global Parameter
Eq. (3)
Eq. (5)
Alignment in Malicious Null Space
Attack Performance
Parameter Space
Keep
Attack Performance
Parameter Space
Eq. (6)
Projection Δ
Null Space
SVD
Singular Value
Eq. (4)
Distribute
Aggregate
Benign Client
Benign Client
Global model
Alignment
Projection
Training Flow
Parameter Flow